BIOSTATISTICS · MIXED EFFECTS · NESTED DESIGN

Statistical Analysis of *Nested Mixed Models* with R

A complete workflow for hierarchically structured biological data — from variance decomposition and model specification to diagnostic testing, power analysis, and post-hoc inference.

R

lme4 · lmerTest
emmeans · DHARMa
performance · car
influence.ME

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01 — EXPERIMENTAL DESIGN

Rats Glycogen Experiment

STRUCTURE

- **TRT** — **Treatment**: Fixed · 3 levels: T1, T2, T3
  - **RAT** — **Rat**: Random · 2 per Treatment · 6 total
    - **LVR** — **Liver Part**: Random · Left / Middle / Right
      - **ε** — **Measurement Replicate**: Residual · 2 per Liver part

N = 3 × 2 × 3 × 2 = **36** observations

$Y_{ijkl} = \mu + \alpha_i + b_{j(i)} + c_{k(ij)} + \varepsilon_{ijkl}$

Experimental Factors

FACTOR	TYPE	LEVELS	DESCRIPTION
Treatment	Fixed	3	Treat1, Treat2, Treat3
Rat	Random	6	Nested within Treatment
Liver	Random	3	Left / Middle / Right
Replicate	Residual	2	Measurement replicates

Fixed vs. Random Effects

Fixed: Specific levels of interest — compare Treatment groups directly.
Random: Sample from a population — account for Rat & Liver variability.

02 — MODEL SPECIFICATION

✓ CORRECT MODEL

Nested Random Effects with Explicit Nesting

```
# ModelR — best specification
lmer(Glycogen ~ Treatment +
  (1 | Treatment:Rat) +
  (1 | Treatment:Rat:Liver),
  REML = TRUE, data = Rats)
```

Treatment is a **fixed** effect. Rat and Liver are nested via explicit colon notation, essential when Rat IDs are non-unique across treatments.

✗ WRONG MODEL 1

Treatment Misspecified as Random Effect

```
# ModelR_wrong — INCORRECT
lmer(Glycogen ~ 1 +
  (1 | Treatment/Rat/Liver),
  REML = TRUE, data = Rats)
# Expands to: (1|Treat) + ...
```

Treatment becomes a **random** effect. No treatment comparison possible. REML = 234.7 vs. 219.6 for correct model — worse fit.

⚠ WRONG MODEL 2

Non-unique Rat IDs Ignoring Treatment Nesting

```
# ModelR_wrong2 — INCORRECT
lmer(Glycogen ~ Treatment +
  (1 | Rat/Liver),
  REML = TRUE, data = Rats)
# Singular fit warning!
```

Assumes Rat IDs are globally unique. Liver variance collapses to ~0. Singular fit. Residual variance = 42.48 (correct model: 21.17).

REML = TRUE
Final model estimation — unbiased variance components

REML = FALSE
Comparing models with different *fixed* effects via AIC / LRT

AIC Lower = Better
Best model: ModelR1 — AIC = 231.62 including both Rat & Liver

03 — MODEL DIAGNOSTICS

Convergence
Gradient check via performance package
✓ CONVERGED

Singularity
Variance not near zero for correct model
✓ NOT SINGULAR

Homoscedasticity
Levene's Test
F(2,33) = 1.45
p = 0.248
✓ p > 0.05

Normality
Shapiro-Wilk
W = 0.976
p = 0.619
✓ p > 0.05

Cook's Distance
All values range 0.000 to 0.025
✓ ALL < 0.5

DHARMa
Simulated residuals uniform QQ looks good
✓ NO VIOLATION

Residuals
Min -1.48
Median 0.03
Max 1.83
✓ SYMMETRIC

check_model()
Comprehensive 6-panel diagnostic suite — passed
✓ ALL PASSED

04 — VARIANCE DECOMPOSITION

Random Effects — Variance Components

Total variance = Rat + Liver + Residual = 71.40 · Rat dominates, confirming biological necessity as random effect

Rat within Treatment (1 | Treatment:Rat)	50.5%	36.06 (SD = 6.005)
Residual Error ε ~ N(0, σ²)	29.6%	21.17 (SD = 4.601)
Liver within Rat (1 | Treatment:Rat:Liver)	19.8%	14.17 (SD = 3.764)

05 — FIXED EFFECTS & ESTIMATED MARGINAL MEANS

Estimated Marginal Means

Adjusted glycogen after accounting for random effects

TREATMENT	EMMEAN	SE	95% CI
Treat 1 (ref)	140	4.71	126–155
Treat 2	151	4.71	136–166
Treat 3	135	4.71	120–150

All CIs overlap substantially → no significant Treatment difference

Model Selection — AIC

ANOVA: Treatment significant (χ^2=6.50, p=0.039)

MODEL	AIC	FIT	NOTE
ModelR1	231.6		Best ★
ModelR2_REML	232.3		–Liver RE
ModelR2	245.9		ML version
ModelR3	248.4		No treatment

06 — POWER ANALYSIS (SIMULATION · N = 100)

Rat Random Effect

Power to detect Liver nested within Rat (LRT simulation)

34% POWER

Very Low Power ✗
Experiment detects the Rat random effect only 34% of the time. Even if Rat variability is real, this study misses it 66% of the time. Results are inconclusive, not evidence of no effect.

POWER vs. 0.80 THRESHOLD
0.34

Treatment Fixed Effect

Power to detect Treatment on Glycogen (LRT simulation)

68% POWER

Low Power ⚠
68% chance of detecting true Treatment effect — below recommended 80%. Non-significant results may reflect insufficient data, not genuine absence of Treatment effect. More rats needed.

POWER vs. 0.80 THRESHOLD
0.68

07 — CONCLUSION

No Significant Treatment Effect Detected

Treatment does not significantly affect Glycogen concentration (F(2,3) = 2.93, p = 0.197). Observed differences between Treat1, Treat2, and Treat3 are not large enough relative to substantial between-rat variability. However, with power below 0.80, this is an inconclusive result — not definitive evidence of no effect.

36 largest variance component — between rats

1. **Rat is the dominant variance source** (σ^2 = 36.06). Biological variability between individual rats must always be modelled as random.
2. **Correct model specification is critical.** Both wrong models produced biased variance estimates, singular fits, or invalid treatment comparisons.
3. **Study is underpowered.** Power = 0.34–0.68. Future experiments need more rats per treatment to achieve reliable statistical inference.
4. **All assumptions satisfied:** Convergence, no singularity, homoscedasticity (p=0.248), normality (W=0.976, p=0.619), no influential observations (Cook's D < 0.5).

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